

# Corollary 6.3 (1)

[Problem. Prove Corollary 6.3.

Proof. By Thm 6.1, the inequalities are true for small enough  $\delta$ , where the threshold depends only on  $N$ .

$$\text{Let } A := \frac{\prod_{j=1}^{N-1} j!}{(2\pi)^{N/2} 2^{N^2/2} \Gamma(N^2/2 + 1)}.$$

Then Thm 6.1 says:

$$\lim_{\delta \rightarrow 0} \frac{\sigma(\{u: \|I - u\| \leq \delta\})}{\delta^{N^2}} = A.$$

By definition, this means that for all  $\varepsilon > 0$ ,  $\exists \psi > 0$  such that if  $0 < |\delta| < \psi$ , then

$$\left| \frac{\sigma(\{u: \|I - u\| \leq \delta\})}{\delta^{N^2}} - A \right|^{(*)} < \varepsilon.$$

(\*) means that

$$- \varepsilon + A < \frac{\sigma(\{u: \|I - u\| \leq \delta\})}{\delta^{N^2}} < \varepsilon + A$$

Multiply by  $\delta^{N^2}$ :

$$\delta^{N^2}(A - \varepsilon) < \sigma(\{u: \|I - u\| \leq \delta\}) < \delta^{N^2}(\varepsilon + A).$$

Choose  $\varepsilon^* > 0$  so that  $A - \varepsilon^* > 0$ , and then set  $C_1 = A - \varepsilon^*$  and  $C_2 = A + \varepsilon^*$ .

Observe that  $\varepsilon^*$  was chosen as a function of  $A$ , which depends only on  $N$ .



Then there exists  $\psi^*(\varepsilon^*)$  such that

for  $0 < \delta < \psi^*$ ,

$$C_1 \delta^{N^2} < \sigma(\{u : \|I - u\| \leq \delta\}) < C_2 \delta^{N^2}.$$

Question: Does  $\psi^*$  depend on anything else besides  $\varepsilon^*$ ?

Couldn't  $\psi^*$  depend on the function  $f$ :

$$f(\cdot) = \frac{\sigma(\{u : \|I - u\| \leq (\cdot)\})}{(\cdot)^{N^2}}?$$

But I suppose  $f$  depends on  $N$ , not on  $\delta$ .

If  $\delta \geq \psi^*(\varepsilon^*)$ , then the upper bound

$$\sigma(\{u : \|I - u\| \leq \delta\}) \leq C_2 \delta^{N^2}$$

is true anyway since Haar measure of any set is bounded by one:

That is,  $C_2$  can be chosen larger (still depending only on  $N$ ). That is,

$$C_2 \delta^{N^2} \geq \underbrace{C_2 (\psi^*)^{N^2}}_{\text{depends only on } N},$$

So just choose  $C_2$  so that

$$C_2 (\psi^*)^{N^2} \geq 1 \geq \sigma(\{u : \|I - u\| \leq \delta\}).$$

To generalize the lower bound to all  $\delta \in (0, \sqrt{N})$ , simply observe that the Haar measure of the ball of radius  $\delta$  increases with  $\delta$ .

$$\sigma(\{u : \|I - u\| \leq \psi^*\}) \leq \sigma(\{u : \|I - u\| \leq \delta\}) \text{ when } \delta \geq \psi^*.$$

$$C_1 \delta^{N^2} \leq (\sqrt{N})^{N^2}, \text{ because } \delta \in [\psi^*, \sqrt{N}).$$

$$\text{Choose } 0 < C_1 < \frac{\sigma(\{u : \|I - u\| \leq \psi^*\})}{(\sqrt{N})^{N^2}}.$$



# Notes from section 6 Challenge 2016

$$\lim_{\delta \rightarrow 0} \frac{\sigma(\{u: \|I-u\| \leq \delta\})}{\delta^{N^2}} = \lim_{\gamma \rightarrow 0} \frac{\gamma^{-N^2}}{(2\pi)^N N!} \int_{B(\gamma)} e^{-\|I\|^2/2}$$

$$\prod_{1 \leq j < k \leq N} (\xi_j - \xi_k)^2 d\xi_1 d\xi_2 \dots d\xi_N.$$

Lemma 6.2:

$$\lim_{\delta \rightarrow 0} \frac{\sigma(\{u: \|I-u\| \leq \delta\})}{\delta^{N^2}} = \frac{1}{(2\pi)^N N!} \int_{B(1)} \prod_{1 \leq j < k \leq N} (\eta_j - \eta_k)^2 d\eta_1 \dots d\eta_N$$

↑  
Euclidean ball of radius 1  
above origin in  $\mathbb{R}^N$ .

Now suppose that  $\psi(x) = \begin{cases} 1 & \text{if } x \in [0, 1] \\ 0 & \text{if } x \notin [0, 1] \end{cases}$

"If the above formula holds for this  $\psi$ , then this is the same as the claim of the lemma!" Why?

$$\psi\left(\frac{h(u)}{\delta^2}\right) = \psi\left(\frac{\|I-u\|^2}{\delta^2}\right)$$

Show that

$$\lim_{\delta \rightarrow 0} \delta^{-N^2} \int_{u(N)} \psi(h(u)/\delta^2) d\sigma(u) = \lim_{\delta \rightarrow 0} \frac{\sigma(\{u: \|I-u\| \leq \delta\})}{\delta^{N^2}}$$

ie, does  $\sigma(\{u: \|I-u\| \leq \delta\}) = \int_{u(N)} \psi\left(\frac{h(u)}{\delta^2}\right) d\sigma(u)$ ?

$$\sigma(\{u: \|I-u\| \leq \delta\}) = \int_{u(N)} 1_{\{u: \|I-u\| \leq \delta\}}(u) d\sigma(u)$$

Does

$$1_{\{u: \|I-u\| \leq \delta\}}(u) = \psi\left(\frac{\|I-u\|^2}{\delta^2}\right)? \quad \text{yes.}$$

Case 1: Suppose  $\|I-u\| \leq \delta$ . Then  $\|I-u\|^2/\delta^2 \leq 1$ , so both sides are = 1.

Case 2: else,  $\checkmark$

□

$$h(u) = f(\lambda_1, \dots, \lambda_N)$$

$$\|I - u\|^2$$

$$2 \sum_{j=1}^N (1 - \operatorname{Re}(\lambda_j)).$$

where  $\lambda_j$  are the eigenvalues of  $u$ .

For any smooth  $\psi: [0, \infty) \rightarrow \mathbb{R}$  with compact support,

$$\int_{u(N)} \psi\left(\frac{h(u)}{g^2}\right) d\sigma(u)$$

$$= \frac{1}{(2\pi)^N N!} \int_{[-\pi, \pi]^N} \psi\left(\frac{2}{g^2} \sum_{j=1}^N (1 - \cos \theta_j)\right) \prod_{1 \leq j, k \leq N} |e^{i\theta_j} - e^{i\theta_k}|^2 d\theta_1 \dots d\theta_N.$$

How does this follow from Weyl's formula?

$$\text{Put } h^1(u) := \psi\left(\frac{h(u)}{g^2}\right).$$

$$\begin{aligned} \text{Then } h^1(u) &= \psi\left(\frac{f(\lambda_1, \dots, \lambda_N)}{g^2}\right) = \psi\left(\frac{2}{g^2} \sum_{j=1}^N (1 - \operatorname{Re}(\lambda_j))\right) \\ &= f^1(\lambda_1, \dots, \lambda_N). \end{aligned}$$

We see that  $f^1$  is indeed "some measurable function of  $N$  complex variables that is symmetric under permutation of coordinates."

Then by Weyl:

$$\begin{aligned} \int_{u(N)} \psi\left(\frac{h(u)}{g^2}\right) d\sigma(u) &= \int_{u(N)} h^1(u) d\sigma(u) \\ &= \frac{1}{(2\pi)^N N!} \int_{[-\pi, \pi]^N} f^1(e^{i\theta_1}, \dots, e^{i\theta_N}) \prod_{1 \leq j, k \leq N} |e^{i\theta_j} - e^{i\theta_k}|^2 d\theta_1 \dots d\theta_N \\ &= \frac{1}{(2\pi)^N N!} \int_{[-\pi, \pi]^N} \psi\left(\frac{2}{g^2} \sum_{j=1}^N (1 - \cos \theta_j)\right) \prod_{1 \leq j, k \leq N} |e^{i\theta_j} - e^{i\theta_k}|^2 d\theta_1 \dots d\theta_N. \end{aligned}$$



$$\eta_i = \theta_i / \delta$$

Notes from Section 6. Challenge 2016

$$\begin{aligned} \lim_{\delta \rightarrow 0} \psi \left( 2 \sum_{j=1}^N \frac{1 - \cos(\delta \eta_j)}{\delta^2} \right) \\ = \psi \left( 2 \sum_{j=1}^N \lim_{\delta \rightarrow 0} \frac{1 - \cos(\delta \eta_j)}{\delta^2} \right) \\ = \psi \left( \sum_{j=1}^N \eta_j^2 \right) \quad (\text{why?}) \end{aligned}$$

$$\text{Does } \lim_{\delta \rightarrow 0} \frac{1 - \cos(\delta \eta_j)}{\delta^2} = \eta_j^2?$$

$$\text{L'Hopital: } \begin{aligned} 1 - \cos(\delta \eta_j) &\rightarrow 1 - 1 = 0 \text{ as } \delta \rightarrow 0. \\ \delta^2 &\rightarrow 0 \text{ as } \delta \rightarrow 0. \end{aligned}$$

$$\frac{d}{d\delta} (1 - \cos(\delta \eta_j)) = \sin(\delta \eta_j) \cdot \eta_j$$

$$\frac{d}{d\delta} (\delta^2) = 2\delta$$

$$\begin{aligned} \Rightarrow \lim_{\delta \rightarrow 0} \frac{1 - \cos(\delta \eta_j)}{\delta^2} &= \lim_{\delta \rightarrow 0} \frac{(\eta_j) (\overset{0 \text{ as } \delta \rightarrow 0}{\sin \delta \eta_j})}{\underset{0 \text{ as } \delta \rightarrow 0}{2\delta}} \\ &= \eta_j^2 / 2. \end{aligned}$$

$$\Rightarrow \lim_{\delta \rightarrow 0} \psi \left( 2 \sum_{j=1}^N \frac{1 - \cos(\delta \eta_j)}{\delta^2} \right) = \psi \left( \sum_{j=1}^N \eta_j^2 \right).$$

So the smoothness of  $\psi$  could have just been continuity of  $\psi$ , to pass the limit inside of  $\psi$ .

$$\sigma(\{u : \|E - u\| \leq \delta\}) = \int_{\mathcal{U}(N)} 1_{\{u : \|E - u\| \leq \delta\}}(u) d\sigma(u)$$

$$= \int_{\mathcal{U}(N)} 1_{[0,1]} \left( \frac{\|E - u\|^2}{\delta^2} \right) d\sigma(u)$$

$$= \frac{1}{(2\pi)^N N!} \int_{[-\pi, \pi]^N}$$

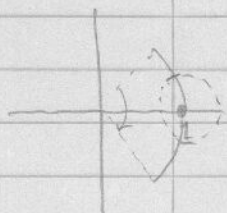
$$1_{[0,1]} \left( \frac{2}{\delta^2} \sum_{j=1}^N (1 - \cos \theta_j) \right) \prod_{\substack{1 \leq j \\ 1 \leq k \leq N}} |e^{i\theta_j} - e^{i\theta_k}|^2 d\theta_1 \cdots d\theta_N.$$

Consider  $U(1)$  or  $N=1$ .

Then

$$\sigma(\{u: \|I-u\| \leq \delta\}) = \frac{1}{2\pi} \int_{[-\pi, \pi]} \mathbb{1}_{[0,1]} \left( \frac{2}{\delta^2} (1 - \cos \theta) \right) d\theta$$

For which  $\theta \in [-\pi, \pi]$  is  $\frac{2}{\delta^2} (1 - \cos \theta) \in [0, 1]$ ?

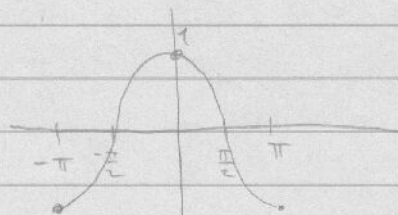


$$0 \leq \frac{2}{\delta^2} (1 - \cos \theta) \leq 1$$

$$0 \leq 2(1 - \cos \theta) \leq \delta^2$$

$$0 \leq (1 - \cos \theta) \leq \frac{\delta^2}{2}$$

$\cos \theta$ :



Let's use also that  $\delta^2 = \frac{1}{8\beta}$ .

$$\rightarrow 0 \leq (1 - \cos \theta) \leq \frac{1}{16\beta}$$

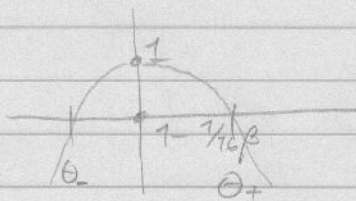
$$\rightarrow \boxed{\cos \theta \leq 1} \leq \frac{1}{16\beta} + \cos \theta$$

always satisfied

Require:  $1 - \frac{1}{16\beta} \leq \cos \theta$ .

$\beta$  is large, so  $1 - \frac{1}{16\beta}$  is close to 1.

So  $\theta$  must be in a neighborhood of 0.



Hence, for  $\delta^2 = \frac{1}{8\beta}$ , we have

$$\sigma(\{u: \|I-u\| \leq \frac{1}{\sqrt{8\beta}}\}) = \frac{1}{2\pi} \int_{[-\pi, \pi]} \mathbb{1}_{[0,1]} \left( \frac{2}{\delta^2} (1 - \cos \theta) \right) d\theta$$

$$= \frac{1}{2\pi} \int_{\{\theta: 1 - \frac{1}{16\beta} \leq \cos \theta\}} 1 d\theta$$

$$= \frac{1}{2\pi} [\theta_{hi} - \theta_{lo}], \text{ where } \theta \approx \cos^{-1}(1 - \frac{1}{16\beta}).$$



Then for any  $n$ , and for  $U(1)$  ( $N=1$ ),

$$Z(B_n, g_0) \geq e^{-|B_n|} \sigma_{B_n}(\{u: S_{B_n}(u) \leq |B_n|/\beta\})$$

$$\geq e^{-|B_n|} \sigma_{B_n}(\{u: \|I - u(x, y)\| \leq \delta\})$$

$\delta = \frac{1}{\sqrt{8\beta}}$

$$= e^{-|B_n|} \prod_{e \in E_n} \left( \frac{1}{2\pi} \int_{\theta_{e1} - \theta_{e0}}^{\theta_{e1} + \theta_{e0}} \right)$$

$\uparrow$  depends on  $\beta$

$\hookrightarrow$  this is something that I can get explicit numbers with, if I take  $\beta$  to be an explicit number and  $n = 7$ , for instance.

Say  
 $\beta = 100$

that is the next step.

Model parameters:  $d=4$ ,  $N=1$ ,  $n=(10, 100, 1000)$ ,  $\beta =$

$$\delta \in (0, \sqrt{N}) = (0, 1)$$

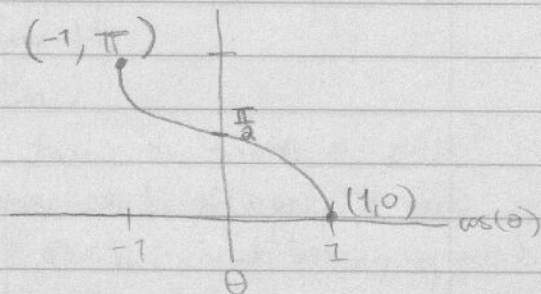
$$\delta = \frac{1}{\sqrt{8\beta}} \Rightarrow \beta = \frac{1}{8\delta^2} \in \left(\frac{1}{8}, \infty\right)$$

Suppose  $\beta = \frac{1}{2}$ .

What is the set  $\{\theta \in [-\pi, \pi] : 1 - \frac{1}{16\beta} \leq \cos \theta\}$ ?

Then the condition is  $1 - \frac{2}{16} = \frac{7}{8} \leq \cos \theta$ . What is the definition of arccos?

$\arccos: [-1, 1] \rightarrow [0, \pi]$



$\arccos(\frac{7}{8})$  is a small positive angle, and we can conclude that  $\theta \in (-\arccos(\frac{7}{8}), \arccos(\frac{7}{8}))$  if and only if  $\frac{7}{8} \leq \cos \theta$  and  $\theta \in [-\pi, \pi]$ .

For arbitrary  $\beta \in (\frac{1}{8}, \infty)$ , we see that  
 $\{\theta \in [-\pi, \pi] \mid 1 - \frac{1}{16\beta} \leq \cos \theta\}$   
 $= (-\arccos(1 - \frac{1}{16\beta}), \arccos(1 - \frac{1}{16\beta}))$ ,

which implies by a previous calculation that

$$\sigma(\{u: \|I-u\| \leq \frac{1}{\sqrt{8\beta}}\}) = \frac{1}{2\pi} (2 \arccos(1 - \frac{1}{16\beta}))$$

$$= \frac{\arccos(1 - \frac{1}{16\beta})}{\pi}.$$

| $\beta$       | $\sigma(\{u: \ I-u\  \leq \frac{1}{\sqrt{8\beta}}\})$ |
|---------------|-------------------------------------------------------|
| $\frac{1}{8}$ | $0.33333 = \frac{1}{3}$                               |
| 1             | 0.77373                                               |
| 10            | 0.0356                                                |

Thm 7.1 is essentially the claim that

$$Z(B_n, g_0) = Z(B_n, \frac{1}{\sqrt{\beta}}) \geq e^{-|B_n'| \left( \frac{\arccos(1 - \frac{1}{16\beta})}{\pi} \right) |E_n|}$$

What is  $|B_n'|$ ? This is the number of plaquettes.

I previously computed that the number of possible plaquette first entries is  $n^d - dn + d - 1$ , but this is not the same as  $|B_n'|$ , because a plaquette first entry can appear in multiple plaquettes, i.e.  $(0, 1, 2)$  and  $(0, 2, 3)$ .

$$B_n' = \{ (x, j, k) \mid x \in B_n, 1 \leq j < k \leq d, \text{ and the resulting plaquette } \checkmark \text{ belongs to } B_n \}$$

vertices all

This is hard to count.

Fix a plaquette base vertex, say,  $0 \in B_n$ . How many plaquettes begin at 0? There are  $d-1$  possible choices for  $j$ . The number of choices for  $k$  depends on the number of choices for  $j$ .

If  $j=1$ , then the options for  $k$  are  $2, \dots, d$ , or  $d-1 = d-j$  total options. Continuing to  $j = d-1$ , we see there are

$$\sum_{j=1}^{d-1} (d-j) \text{ possible } (j, k) \text{ pairs.}$$



For instance, in the case  $d=5$ , then there are  $4+3+2+1=10$  options for  $(j,k)$ , where  $1 \leq j < k \leq 5$ . We have

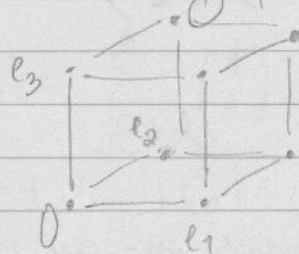
$$\begin{aligned} \sum_{j=1}^{d-1} (d-j) &= (d-1)(d) - \sum_{j=1}^{d-1} j \\ &= (d-1)(d) - \frac{(d)(d-1)}{2} \\ &= \frac{d(d-1)}{2} = \frac{d^2-d}{2} \end{aligned} \quad \begin{aligned} \sum_{j=1}^3 j &= \frac{(4)(3)}{2} \\ &= 6 \checkmark \end{aligned}$$

So there are  $\frac{d^2-d}{2}$  plaquettes beginning with vertex 0.



$$d=2 \rightarrow \frac{2^2-2}{2} = \frac{4-2}{2} = 1 \quad \text{verified } \checkmark$$

But consider boundary plaquettes when  $d \geq 3$ :



Plaquettes beginning with  $e_1$ :

$$(e_1, 2, 3)$$

Plaquettes beginning with  $e_2$ :

$$(e_2, 1, 3)$$

Plaquettes beginning with  $e_3$ :

$$(e_3, 1, 2)$$

Plaquettes beginning with 0:

$$(0, 1, 2)$$

$$(0, 1, 3)$$

$$(0, 2, 3)$$

So if a base vertex is already maxed out in one or more entries, then we must count differently.

Fix  $n$  and  $d$ . Let  $x \in B_n$ .

$$x = (x_1, \dots, x_d)$$

$$x_i \in \{0, \dots, n-1\} \forall i \in [d]$$

Each entry  $i \in [d]$  is either maxed out (i.e.,  $x_i = n-1$ ) or it is not (a binary decision). At most  $d$  entries are maxed out, and at least 0 entries are maxed out. If  $k$  entries are maxed out, there are  $\binom{d}{k}$  ways for this to happen.

How many plaquettes are possible when  $k$  entries are maxed out? This reduces to the case of the origin 0 when the dimension is  $d-k$ . So there are  $\frac{(d-k)(d-k-1)}{2}$  possibilities.

Observe that if  $d-1$  or  $d$  entries are mixed out, then there are no plaquettes. For example,  $(e_1 + e_2) \in B_2$  for  $d=3$  (see picture on previous page) has no plaquettes. Thus, the total number of boundary plaquettes is

$$(n-1)^{d-1} \binom{d}{1} \frac{(d-1)(d-1-1)}{2} + \dots + (n-1)^2 \binom{d}{d-2} \frac{(d-(d-2))(d-(d-2)-1)}{2}.$$

because when  $k$  entries are mixed out (i.e.  $= n-1$ ), then all other entries have  $n-1$  options.

This formula looks nasty, but we can just look at  $d=3, 4$ . Then it is simple:

$$(n-1)^3 \binom{4}{1} \frac{3 \cdot 2}{2} + (n-1)^2 \binom{4}{2} \quad \text{for } d=4$$

$$(n-1)^2 \binom{3}{1} \quad \text{for } d=3.$$

Observe that  $(n-1)^2 \binom{3}{1}$  agrees with the previous page where  $d=3$  and  $n=2$ .

How many non-boundary plaquettes are there?

There are  $(n-1)^d \cdot \frac{(d)(d-1)}{2}$  of them,

because there are  $n-1$  choices for each entry of the vertex  $x$  in  $(x, j, k)$ , and then  $\frac{(d)(d-1)}{2}$  choices for  $(j, k)$ .

Also, a clarification is that  $(0, 1, 2)$  could perhaps be viewed as a boundary plaquette but I mean so mean that none of the entries of the base vertex are mixed out.

So



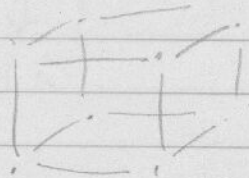
Total plaquettes for  $d=4$ ,  $n$  given:

$$(n-1)^4 \frac{(4)(3)}{2} + (n-1)^3 \binom{4}{1} \frac{3-2}{2} + (n-1)^2 \binom{4}{2}$$

Total plaquettes for  $d=3$ ,  $n$  given:

$$(n-1)^3 \frac{(3)(2)}{2} + (n-1)^2 \binom{3}{1}$$

For  $n=2$  and  $d=3$ :



$$(1)^3 \frac{(3)(2)}{(2)} + (1) \binom{3}{1} = 3+3$$

Agrees with the number of plaquettes.

What is  $|E_n|$ ? Chatterjee computed this:

$$|E_n| = \sum_{l=1}^d \binom{d}{l} l (n-1)^l = d n^{d-1} (n-1).$$

For  $d=4$ ,  $|E_n| = 4n^3(n-1).$

For  $d=4$ :  $Z(B_n, g_0) \geq$

$$\exp(-(n-1)^4 \cdot 6 - (n-1)^3 \cdot 12 - (n-1)^2 \cdot 6) \\ \times \left( \frac{\arccos\left(1 - \frac{1}{16\beta}\right)}{\pi} \right)^{4n^3(n-1)} \quad \text{when } N=1.$$

For  $\beta=1$  and  $n=10$ , this is:

$$Z(B_{10}, 1) \geq \exp(-48600) \times \left( 0.113134 \right)^{4(1000)(9)} \\ \approx 0.$$